# SAQIB FARAZ

Hardware Development Engineer | Embedded Firmware & PCB Design | Sensor Signal Measurement
+49 17671229700 | saqibfrz@gmail.com | Wismar, Germany | linkedin.com/in/saqibfrz | Portfolio: tinyurl.com/saqibfrz

## PROFESSIONAL OVERVIEW

M.Eng. candidate in Information & Electrical Engineering with hands-on experience in mixed-signal PCB development, sensor interfacing, and embedded measurement systems. Designed, assembled, and validated current, voltage, temperature, and DAC boards for a hydrogen fuel-cell vehicle, including KiCad-based multilayer PCB layout, component selection, board bring-up, and lab debugging. Developed ESP32 firmware for SPI-based data acquisition, interrupts, PWM-triggered sampling, and data averaging, and performed gain, scaling, LSB, and error analysis during validation. Comfortable with oscilloscope, logic analyzer, DMM, bench PSU, and function generator. Work-authorized in Germany; open to relocation.

## TECHNICAL SKILLS

- **PCB Design:** KiCad (schematic + layout, 2–6 layers), Altium Designer (familiarity), ERC/DRC, Gerbers, BOM, assembly notes, component selection, sensor signal conditioning, analog front-end design basics, DFM, grounding, decoupling, layer stack-up
- **Hardware Bring-up & Validation:** SMD assembly, oscilloscope, logic analyzer, DMM, bench PSU, function generator, rail verification, SPI timing validation, signal integrity checks, noise/error debugging, measured vs. theoretical comparison
- **Embedded Firmware:** Embedded C (ESP32), SPI, I²C, UART, ADC interfacing, PWM, timers, interrupts, state machines, data averaging, digital filtering, PlatformIO, VS Code, Git
- **Simulation & Analysis:** LTspice, MATLAB/Simulink, gain/scaling/filter calculations, LSB and error analysis, Python (basic), FreeCAD (basic), MS Office
- **Additional:** FPGA/VHDL, Xilinx Vivado, simulation, testbench (academic project experience)

## PROJECT EXPERIENCE

**Hydrogen Fuel Cell Vehicle — Shell Eco-Marathon** *Feb 2024 – Present*

*Hochschule Wismar, Germany | Master's Thesis (registered Feb 2026, expected Sep 2026): Development and Integration of Measurement & Control Modules for Reliable Operation in a Fuel-Cell Vehicle*

- Designed, assembled, and validated custom current, voltage, temperature, and DAC boards for a hydrogen fuel-cell vehicle, including schematic capture, KiCad-based multilayer PCB layout, component selection, and board bring-up.
- Integrated and tested mixed-signal components including ADS7229, LTC2368, AD8410A, MAX49925, precision references, LDOs, and op-amps for sensor interfacing and signal conditioning.
- Developed ESP32 firmware for SPI-based acquisition, interrupt/timer-driven control, PWM-triggered sampling, and data averaging/filtering in PlatformIO/VS Code.
- Validated an ADS7229 voltage-measurement chain across a 22-point sweep from 0–4.0 V, achieving ~2.5 mV mean absolute error and ~5.7 mV max error over the main operating range, with <1 code standard deviation in repeatability tests.
- Characterized a CT425 current-sensing signal chain across 48 test points up to 9.4 A and compared alternative front ends including AD8410A and MAX49925 to support revision decisions.
- Performed gain, scaling, filter, and LSB calculations and used measured-vs-theoretical comparisons to diagnose grounding issues, timing mismatches, noisy rails, offset/scaling errors, and op-amp configuration problems during validation.

**FPGA-Based Digital Stopwatch** *Sep 2023 – Jan 2024*

*Hochschule Wismar, Germany*

- Designed and verified a VHDL-based digital stopwatch on FPGA with 0.1 s resolution, including timing/control logic and 7-segment display driving; validated using simulation and testbench in Xilinx Vivado.

**Power Measurement Board** *Mar 2023 – Jul 2023*

*Hochschule Wismar, Germany*

- Designed and validated an analog front-end and ADC interface for a power-measurement board, developed embedded C firmware with digital filtering, and verified signals using oscilloscope and logic analyzer.

## ADDITIONAL EXPERIENCE

**Executive Assistant | Super Electronics and Battery Works, Jaunpur, India** *May 2019 – May 2022*

- Managed operations, finance reporting, supplier follow-up, and process coordination for an electronics and battery business; improved working-capital tracking using Excel-based reporting.

## EDUCATION

**M.Eng. Information & Electrical Engineering | Hochschule Wismar, Germany** *Sep 2022 – Expected Sep 2026*

**B.Tech. Instrumentation Engineering | VBS Purvanchal University, India** *Sep 2014 – Jul 2018*

## LANGUAGES

English (C1 — Professional) | German (B1 certified; B2 in preparation) | Hindi (Native)